

Extracellular acidification assay to evaluate the effectiveness of antifungal agents on the pathogenicity of *Macrophomina phaseolina*

Krishna R Acharya¹, Aswin Reddy Chilakala¹ and Muthappa Senthil-Kumar^{1*}

¹ National Institute of Plant Genome Research, New Delhi, India, 110067.

*Corresponding author

Muthappa Senthil-Kumar

National Institute of Plant Genome Research, New Delhi, India, 110067.

Tel- 91-11- 26735229,

Fax- 91-11-26741658

E-mail- skmuthappa@nipgr.ac.in

<http://orcid.org/0000-0003-1502-1659>

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Abstract

Macrophomina phaseolina is a fungus that causes dry root rot disease, causing considerable yield loss worldwide. Fungi exhibit various ways of absorbing nutrients through their plasma membrane, such as free or facilitated diffusion, diffusion channels, or active transport. Glucose, as a preferred carbon source, activates the plasma membrane H⁺-ATPase, resulting in the release of protons. Consequently, the protons, along with the organic acid metabolites released into the extracellular environment, acidify the cell surroundings. This decrease in pH cues the fungus to shift from saprotrophic to necrotrophic growth, facilitating host invasion. Sustainable dry root rot disease management often relies on the employment of antifungal agents from various biological sources. Despite the discovery of numerous antifungal agents, only a limited number have been evaluated for their efficacy against this phytopathogenic fungus. This scarcity of testing is primarily due to the limitations of existing methods, which often lack standardisation and reproducibility. This chapter introduces a rapid and sensitive method to assess the antifungal activity of various agents against *M. phaseolina*. By measuring extracellular pH changes after treatment in the presence of a nutrient source, we can determine the inhibitory concentrations of these agents and evaluate their potential for controlling fungal pathogenicity in plants.

Keywords: *Macrophomina phaseolina*, extracellular acidification, phytopathogenesis, microsclerotia, fungal physiology, antifungal agent screening, pH regulation.

1. Introduction

1.1 Plasma membrane physiology

In fungal physiology, membrane transport systems are crucial because they control effective cell reproduction, development, and metabolism. The selectively permeable membrane allows nutrient absorption and release of metabolites from the fungal cell. Fungi exhibit several active and passive absorption mechanisms at the plasma membrane viz., free diffusion, facilitated diffusion, diffusion channels, and active transport. In filamentous fungi, active transport of nutrients including sugars, amino acids, nitrate, ammonium, sulphate, and phosphate demands spatial separation of the ion pumps, which are mostly located behind the apex while the symport proteins are active near the tip. Thus, when the hyphal tip continually digs into new resources, it absorbs nutrients, and the mitochondria located behind the apex provide ATP to fuel the ion pump and produce proton motive force [1].

1.2 Cellular pH regulation and glucose uptake

In almost all cells, pH control is mainly achieved through the combined activity of pumps, exchangers, and buffers, supported by multiple layers of regulation [2]. **Fig. 1** shows proton pumps, such as Pma1 (P-type H⁺-ATPase) at the fungal plasma membrane, which determine membrane potential by electrogenic transport of H⁺ without counterions and highlights V-ATPases on fungal organelle membranes, which couple ATP hydrolysis with proton transport to establish pH gradients [3]. ATP-binding cassette (ABC) transporters protect *M. phaseolina* from host-derived secondary metabolites and supply essential nutrients [4].

Under conditions of abundant glucose, their preferred carbon source, *S. cerevisiae* cells will undergo rapid fermentative growth, producing ethanol, CO₂, and organic

acids [5, 6]. Cells grown in glucose rapidly acidify their medium and require robust mechanisms to maintain cytosolic pH during growth, and cytosolic pH decreases as cells reach the stationary phase [3]. *Aspergillus niger* induces the expression of pH-dependent monosaccharide/H⁺ symporter upon carbon starvation [7]. The growth of *Penicillium ochrochloron* under glucose-limited conditions allowed for immediate glucose intake at higher rates from a glucose pulse compared to steady-state conditions. [8]. In addition to the efflux of acidic metabolites, activation of plasma membrane H⁺-ATPase by glucose contributes to about one-half of extracellular acidification [9]. Inhibition of plasma membrane Pma1 (H⁺-ATPase) disrupts the electrochemical gradient, nutrient transporters and extracellular acidification in many fungi [10].

1.3 Phytopathogenic aspects of extracellular acidification

Pathogenic fungi such as *Sclerotinia* and *Botrytis* sp. produce an acidic environment that causes a shift from saprotrophic to necrotrophic growth in plants [11]. In *Fusarium* sp, fusaric acid acts as a virulence factor during the early stages of the disease [12]. Growth of *M. phaseolina* in the presence of host tissue triggers azelaic acid production, potentially aiding in disease establishment [13]. Endoglucanases catalysing the hydrolysis of cellulose to glucose, and malate dehydrogenase causing organic acid production, were found in the secretome study of *M. phaseolina* [14]. Additionally, several investigations demonstrated the significance of plasma membrane H⁺-ATPases during infection and pathogenesis [15, 16].

1.4 Dry root rot disease in chickpea

M. phaseolina propagates through dark-coloured, branching hyphae with black microsclerotia [13]. Infected roots when split exhibit characteristic dark spots in the

infected regions, which are microsclerotia and serve as the best parameter to identify the disease under the field conditions [17]. Several surveys and field investigations indicate that drought and high temperatures increase the prevalence and severity of DRR disease [17–21]. Moreover, they usually occur as mixed infections usually with wilt [22]. Dry root rot (DRR) of chickpea had its impact over the 13 million hectares of chickpea-growing regions worldwide leading to a significant reduction in yield of 30 to 50 per cent under moderate infection, and up to 80 per cent yield reductions in susceptible genotypes [17, 18, 21, 23, 24].

High resistance levels are challenging to establish because of the pathogen's wide host range [25]. None of the lines were fully resistant to DRR, according to a search for particular resistance in chickpea [26]. Therefore, the reliability of the effective sustainable metabolites and proteins increases in disease control. Although numerous fungi, bacteria, and plants have been found to contain antifungal metabolites/proteins that are effective phytopathogenic fungi, their effective characterization is limited by diverse testing methods usually lacking standardized conditions and reproducibility.

1.5 Some common methods used for screening antifungal compounds

The search for natural agents with antifungal activity has increased, but testing methods present many challenges. Techniques like agar diffusion, macro dilution, and microdilution are used to determine antifungal activity. Agar diffusion is semi-quantitative and involves applying a sample to an agar surface inoculated with fungal cells. Macro dilution and microdilution provide quantitative data about a compound's Minimum Inhibitory Concentration. Colorimetric assays, such as MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide) and XTT (2,3-bis-(2-methoxy-

4-nitro-5-sulphenyl)-(2H)-tetrazolium-5-carboxanilide), are also used to estimate fungal biomass and metabolic activity.

Obtaining intact *M. phaseolina* inoculum across treatments is difficult due to its nature of propagation in synthetic media, making subsequent impact assessment challenging. Moreover, the existing methods are usually expensive, time-consuming and cause stress to the fungus during sample preparation and processing. This assay tests the effect of an antifungal agent (AFA) on actively grown mycelium using extracellular pH measurements after treatments in the presence of glucose (**Fig. 2**). The pH reductions can be monitored using a laboratory pH meter [27].

2. Materials

2.1 Fungal sample preparation

1. Potato dextrose agar (PDA)
2. 90 mm sterile disposable Petri plates
3. 20 mL test tubes
4. Standard 1 mL micropipette and tips
5. Parafilm
6. 70 % ethanol
7. Pure culture of *M. phaseolina* in PDA
8. Laminar air flow chamber
9. 28°C incubator

2.2 Growth in culture plate

1. Potato dextrose broth (PDB)
2. Flat bottom 12 well culture plate

140 3. Standard 200 µL micropipette and tips

141 4. L-rod

142 5. Micropore tape

143 6. 28°C shaking incubator

144 7. Laminar air flow chamber

145 **2.3 Treatments and AFA to be tested**

146 1. Test AFA (see **Note 1**).

147 2. 0.2 µm filter and 10 mL syringe

148 3. 1 mL micropipette and tip

149 4. Micropore tape

150 5. Positive control and Negative control (see **Note 2**)

151 **2.4 Nutrient source and pH measurement**

152 1. 2 % D – glucose (anhydrous) solution in Milli-Q® (MQ) water

153 2. Autoclaved MQ

154 3. 0.2 µm filter and 10 mL syringe

155 4. Autoclaved 15 mL glass vials

156 5. MQ in wash bottle and tissue paper

157 6. pH meter (see **Note 3**)

158

159 **3. Methods**

160 **3.1 *M. phasolina* agar culture**

161 1. Autoclave the desired number of test tubes with 10 mL of PDA medium (see **Note**

162 **4**).

2. Perform the subsequent steps inside a laminar airflow chamber with all equipment wiped with 70% ethanol.
3. Pour the molten PDA agar into the sterile Petri plates and let it solidify and dry properly.
4. Ensure that the surface of the pure culture on the medium is uniform (see **Note 5**).
5. Pick the microsclerotia each time from a definite radius of the dark-coloured region to ensure uniform inoculum.
6. Stab the pure culture using a 1 mL tip with a micropipette in a direction perpendicular to the agar surface (see **Note 6**).
7. Depress the plunger by keeping the tip close to the centre of the freshly prepared agar plate (see **Note 7**).
8. Seal the Petri plate twice with parafilm to avoid contamination and incubate it at 28°C for 3 days.
9. Perform the above inoculation three times to obtain biological replicates.

3.2 *M. phaseolina* broth culture

1. Perform the subsequent steps inside a laminar airflow chamber with each equipment wiped with 70% ethanol.
2. Follow **steps 4 and 5** from **method 3.1**.
3. Stab the pure culture using a 200 µL tip in a direction perpendicular to the agar surface (see **Note 6**).
4. Depress the plunger to release the micro-agar plug into the wells of the 12-well culture plate (see **Note 7**).
5. After inoculation, pour 3 mL of sterile PDB into each well and wrap it with micropore tape.
6. Incubate the culture plate at 28°C for 24 h at 180 rpm shaking incubator.

3.3 Treatment of *M. phaseolina* with the AFA and controls

1. Perform the subsequent steps inside a laminar airflow chamber with each equipment wiped with 70% ethanol.
2. Select the germinated micro-agar plugs that have similar mycelial morphology.
3. Remove the broth from the culture plate using a 1 mL pipette (see **Note 8**).
4. Wash the mycelia 3 times in the culture plate for 5 minutes at a 180 rpm shaker and suspend them in 3 mL of respective solutions (see **Note 9**).
5. Transfer the mycelia into a new 12-well culture plate using a 1 mL cut tip (see **Note 10**).
6. Add 3 mL each of filter-sterilised test AFA, positive control, and negative control solutions with 3 replicates (see **Note 11**).
7. Seal the culture plate with micropore tape and incubate the culture plate at 28°C for 24 h at 180 rpm shaking incubator.

3.4 Measurement of the glucose-induced acidification

1. Remove the solutions from the culture plate and wash the treated mycelia 3 times with 3 mL of MQ as in **steps 3 and 4 of method 3.3** and suspend them in MQ until further process.
2. Add 10 mL of filter-sterilised glucose solution to the glass vials.
3. Transfer the treated mycelia to the glass vials using a 1 mL cut tip (see **Note 10**).
4. Mix the contents at 180 rpm between measurements.
5. Dip the pH probe into glass vials for the measurements (see **Note 12**).
6. Record the measurements at 0 and 60 minutes, wash the probe with MQ and blot dry it between replicates (see **Note 13**).
7. Repeat **methods 3.2, 3.3 and 3.4** for the biological replicates.

3.5 Determination of the effective AFA concentration

1. Obtain Δ pH values for each treatment and control sample by subtracting the final values from the initial values.
2. Create an XY data table. Enter the concentrations of the AFA into X. Enter the average of Δ pH of replicates into Y.
3. From the data table, click Analyze, choose nonlinear regression, choose the panel of equations "Dose-response curves - Inhibition" and then choose the equation "[Inhibitor] vs. response - Variable slope" (see **Note 14**).
4. Enter the average Δ pH of negative and positive control as top and bottom constraints ("type: constant equal to") values respectively in the option and plot the dose-response curve (**Fig. 3**).
5. Obtain the metadata of IC₅₀ analysis from the result section (see **Note 15**).

3.6 In vivo evaluation of fungal pathogenicity

The fungal sample post-treatments can be used for in-planta experiments by employing the blotting paper method [28].

1. Surface-sterilise the chickpea seeds with 2% sodium hypochlorite.
2. Sow seeds in a 15 cm height pot containing soilrite.
3. Uproot 8-day-old seedlings, wash under running tap water and rinse with sterile water twice.
4. Arrange the plants over the blotter paper in trays and add sufficient water.
5. Transfer an equal number of fungal samples after treatment into each main root of the plants using a 1 mL cut tip.
6. Place the trays with the plants at 28°C for 8 days under a 16 h photoperiod and 70% relative humidity, moistening the plant daily with sufficient water (see **Note 16**).

7. Calculate the disease susceptibility score/disease severity index and analyse the impact of AFA on the pathogenicity of the fungus (see **Note 17**).

4. Notes

1. Solvents without inhibitory effect on the fungus can be used to test multiple AFAs. An antifungal protein dissolved in 1 X PBS was used in this chapter for representation.
2. Positive control: A commercially available AFA that shows maximum inhibition response. Negative control: solvent for dissolving AFA.
3. Use a pH meter with a combination electrode probe and a resolution of 0.01 or more.
4. The uniform agar plate preparation will ensure an equal size of the subsequent micro-agar plug inoculum.
5. If the culture contains aerial mycelial growth, remove it using an L-rod and follow the same step for each subculture.
6. Ensure proper insertion of a single micro-agar plug inside the tip during each transfer.
7. Use a fresh tip whenever there is more than one micro-agar plug inside the tip or it is not being released.
8. Tilt the plate 45° and remove the broth from both sides of the mycelia without touching it. Tilt it to the opposite side and remove the excess broth.

9. Use appropriate solvents to wash the positive control, negative control and test samples.
10. This prevents mycelial damage between transfers. Press the plunger, place the cut tip over the centre of the mycelium, and gently release till the entire mycelium is inside the tip. Press the plunger to transfer.
11. Optimise the range of AFA concentrations and number of replicates for the experiment to obtain a reliable dose-response curve.
12. Ensure that the porous reference junction of the probe is dipped inside the test solution along with the glass bulb before measurement.
13. Select the time points based on experiment needs. Avoid rubbing the sensitive glass bulb of the probe. Use tissue paper to remove the excess water.
14. The statistical analysis was performed using GraphPad Prism version 10.2.0 for Windows, GraphPad Software, Boston, Massachusetts USA, for further details visit www.graphpad.com.
15. The IC₅₀ value denotes the concentration of AFA that elicits 50% of the response.
16. Choose the method of inoculation and day of plant sample collection based on the phytopathogen under study.
17. The AFA concentration that exhibits the optimum inhibition and simultaneously shows the least infection in the plant can be selected for downstream applications.

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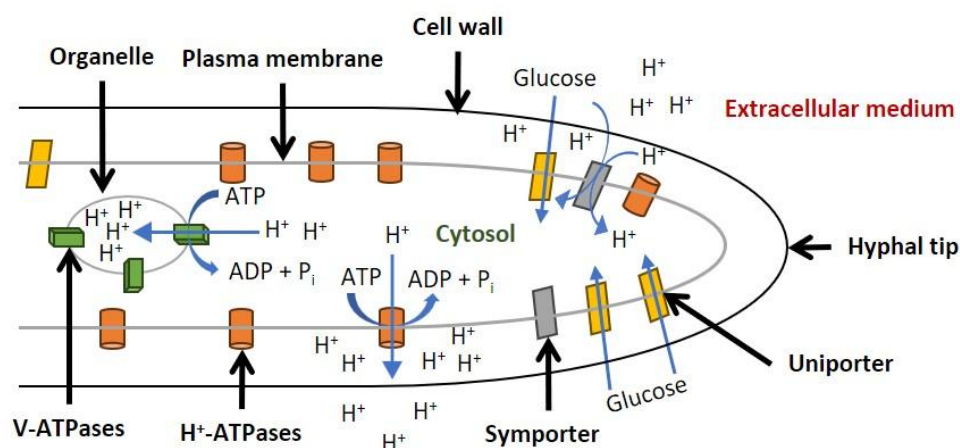
286 **References**

- 287 1. Walker GM, White NA (2017) Introduction to Fungal Physiology. In: Fungi. John
288 Wiley & Sons, Ltd, pp 1–35
- 289 2. Casey JR, Grinstein S, Orlowski J (2010) Sensors and regulators of intracellular
290 pH. *Nat Rev Mol Cell Biol* 11:50–61. <https://doi.org/10.1038/nrm2820>
- 291 3. Kane PM (2016) Proton Transport and pH Control in Fungi. In: Ramos J,
292 Sychrová H, Kschischo M (eds) *Yeast Membrane Transport*. Springer
293 International Publishing, Cham, pp 33–68
- 294 4. Islam MS, Haque MS, Islam MM, et al (2012) Tools to kill: Genome of one of the
295 most destructive plant pathogenic fungi *Macrophomina phaseolina*. *BMC*
296 *Genomics* 13:493. <https://doi.org/10.1186/1471-2164-13-493>
- 297 5. Johnston M (1999) Feasting, fasting and fermenting: glucose sensing in yeast
298 and other cells. *Trends in Genetics* 15:29–33. [https://doi.org/10.1016/S0168-](https://doi.org/10.1016/S0168-9525(98)01637-0)
299 [9525\(98\)01637-0](https://doi.org/10.1016/S0168-9525(98)01637-0)
- 300 6. Conrad M, Schothorst J, Kankipati HN, et al (2014) Nutrient sensing and signaling
301 in the yeast *Saccharomyces cerevisiae*. *FEMS Microbiology Reviews* 38:254–
302 299. <https://doi.org/10.1111/1574-6976.12065>
- 303 7. vanKUYK PA, DIDERICH JA, MacCABE AP, et al (2004) *Aspergillus niger* mstA
304 encodes a high-affinity sugar/H⁺ symporter which is regulated in response to
305 extracellular pH. *Biochemical Journal* 379:375–383.
306 <https://doi.org/10.1042/bj20030624>

8. Vrábl P, Mutschlechner W, Burgstaller W (2008) Characteristics of glucose uptake by glucose- and NH₄-limited grown *Penicillium ochrochloron* at low, medium and high glucose concentration. *Fungal Genetics and Biology* 45:1380–1392. <https://doi.org/10.1016/j.fgb.2008.07.017>
9. Kotyk A, Lapathitis G, Křenková Š (1999) Glucose- and K⁺-induced acidification in different yeast species. *Folia Microbiol* 44:295–298. <https://doi.org/10.1007/BF02818550>
10. Kjellerup L, Gordon S, Cohrt KO, et al (2017) Identification of Antifungal H⁺-ATPase Inhibitors with Effect on Plasma Membrane Potential. *Antimicrobial Agents and Chemotherapy* 61:10.1128/aac.00032-17. <https://doi.org/10.1128/aac.00032-17>
11. Vylkova S (2017) Environmental pH modulation by pathogenic fungi as a strategy to conquer the host. *PLOS Pathogens* 13:e1006149. <https://doi.org/10.1371/journal.ppat.1006149>
12. Liu S, Li J, Zhang Y, et al (2020) Fusaric acid instigates the invasion of banana by *Fusarium oxysporum* f. sp. *cubense* TR4. *New Phytologist* 225:913–929. <https://doi.org/10.1111/nph.16193>
13. Marquez N, Giachero ML, Declerck S, Ducasse DA (2021) *Macrophomina phaseolina*: General Characteristics of Pathogenicity and Methods of Control. *Frontiers in Plant Science* 12:
14. Sinha N, Patra SK, Ghosh S (2022) Secretome Analysis of *Macrophomina phaseolina* Identifies an Array of Putative Virulence Factors Responsible for Charcoal Rot Disease in Plants. *Frontiers in Microbiology* 13:

15. Vázquez-Carrada M, Feldbrügge M, Olicón-Hernández DR, et al (2022) Functional Analysis of the Plasma Membrane H⁺-ATPases of *Ustilago maydis*. *Journal of Fungi* 8:550. <https://doi.org/10.3390/jof8060550>
16. Wu L, Yuan Z, Wang P, et al (2022) The plasma membrane H⁺-ATPase FgPMA1 regulates the development, pathogenicity, and phenamacril sensitivity of *Fusarium graminearum* by interacting with FgMyo-5 and FgBmh2. *Molecular Plant Pathology* 23:489–502. <https://doi.org/10.1111/mpp.13173>
17. Sinha R, Irulappan V, Mohan-Raju B, et al (2019) Impact of drought stress on simultaneously occurring pathogen infection in field-grown chickpea. *Sci Rep* 9:5577. <https://doi.org/10.1038/s41598-019-41463-z>
18. Chilakala AR, Mali KV, Irulappan V, et al (2022) Combined Drought and Heat Stress Influences the Root Water Relation and Determine the Dry Root Rot Disease Development Under Field Conditions: A Study Using Contrasting Chickpea Genotypes. *Front Plant Sci* 13:.. <https://doi.org/10.3389/fpls.2022.890551>
19. Irulappan V, Kandpal M, Saini K, et al (2022) Drought Stress Exacerbates Fungal Colonization and Endodermal Invasion and Dampens Defense Responses to Increase Dry Root Rot in Chickpea. *MPMI* 35:583–591. <https://doi.org/10.1094/MPMI-07-21-0195-FI>
20. Sharma M, Pande S (2013) Unravelling Effects of Temperature and Soil Moisture Stress Response on Development of Dry Root Rot [*Rhizoctonia bataticola* (Taub.)] Butler in Chickpea. *American Journal of Plant Sciences* 4:584–589. <https://doi.org/10.4236/ajps.2013.43076>

- 353 21. Sinha R, Irulappan V, Patil BS, et al (2021) Low soil moisture predisposes field-
354 grown chickpea plants to dry root rot disease: evidence from simulation modeling
355 and correlation analysis. *Sci Rep* 11:6568. [https://doi.org/10.1038/s41598-021-](https://doi.org/10.1038/s41598-021-85928-6)
356 85928-6
- 357 22. Chilakala AR, Pandey P, Durgadevi A, et al (2023) Drought attenuates plant
358 responses to multiple rhizospheric pathogens: A study on a dry root rot-
359 associated disease complex in chickpea fields. *Field Crops Research*
360 298:108965. <https://doi.org/10.1016/j.fcr.2023.108965>
- 361 23. Gupta O, Sharma M (2015) Dry root rot of chickpea: An overview. *Journal of Food*
362 *Legumes*
- 363 24. Rai A, Irulappan V, Senthil-Kumar M (2022) Dry Root Rot of Chickpea: A Disease
364 Favored by Drought. *Plant Disease* 106:346–356. [https://doi.org/10.1094/PDIS-](https://doi.org/10.1094/PDIS-07-21-1410-FE)
365 07-21-1410-FE
- 366 25. Singh KB, Reddy MV (1991) Advances In Disease-Resistance Breeding In
367 Chickpea1. In: Brady NC (ed) *Advances in Agronomy*. Academic Press, pp 191–
368 222
- 369 26. Sharma M, Ghosh R, Pande S (2015) Dry root rot (*Rhizoctonia bataticola* (Taub.)
370 Butler): an emerging disease of chickpea – where do we stand? *Archives of*
371 *Phytopathology and Plant Protection* 48:797–812.
372 <https://doi.org/10.1080/03235408.2016.1140564>
- 373 27. Ghosh P, Roy A, Hess D, et al (2015) Deciphering the mode of action of a mutant
374 *Allium sativum* Leaf Agglutinin (mASAL), a potent antifungal protein on



Rhizoctonia solani. BMC Microbiology 15:237. <https://doi.org/10.1186/s12866-015-0549-7>

28. Irulappan V, Senthil-Kumar M (2021) Dry Root Rot Disease Assays in Chickpea: a Detailed Methodology. JoVE (Journal of Visualized Experiments) e61702. <https://doi.org/10.3791/61702>

Figure 1. Representation of H^+ and glucose transport in active hyphae. The figure shows a section of fungal hyphae with different cell layers and important membrane transporters involved in cellular homeostasis. Upon the recognition of extracellular nutrients (e.g., glucose), the metabolic activity in the growing end of the hyphae is upregulated and causes the secretion of catabolic enzymes. Eventually, the H^+ ions generated with other organic acid by-products create an acidic extracellular environment. This favours the uptake of the nutrients and invasion into host tissues.

Orange cylinders: H^+ -ATPases, yellow parallelograms: glucose uniporter, grey
parallelograms: glucose/ H^+ symporter, green cuboids: V-type ATPases

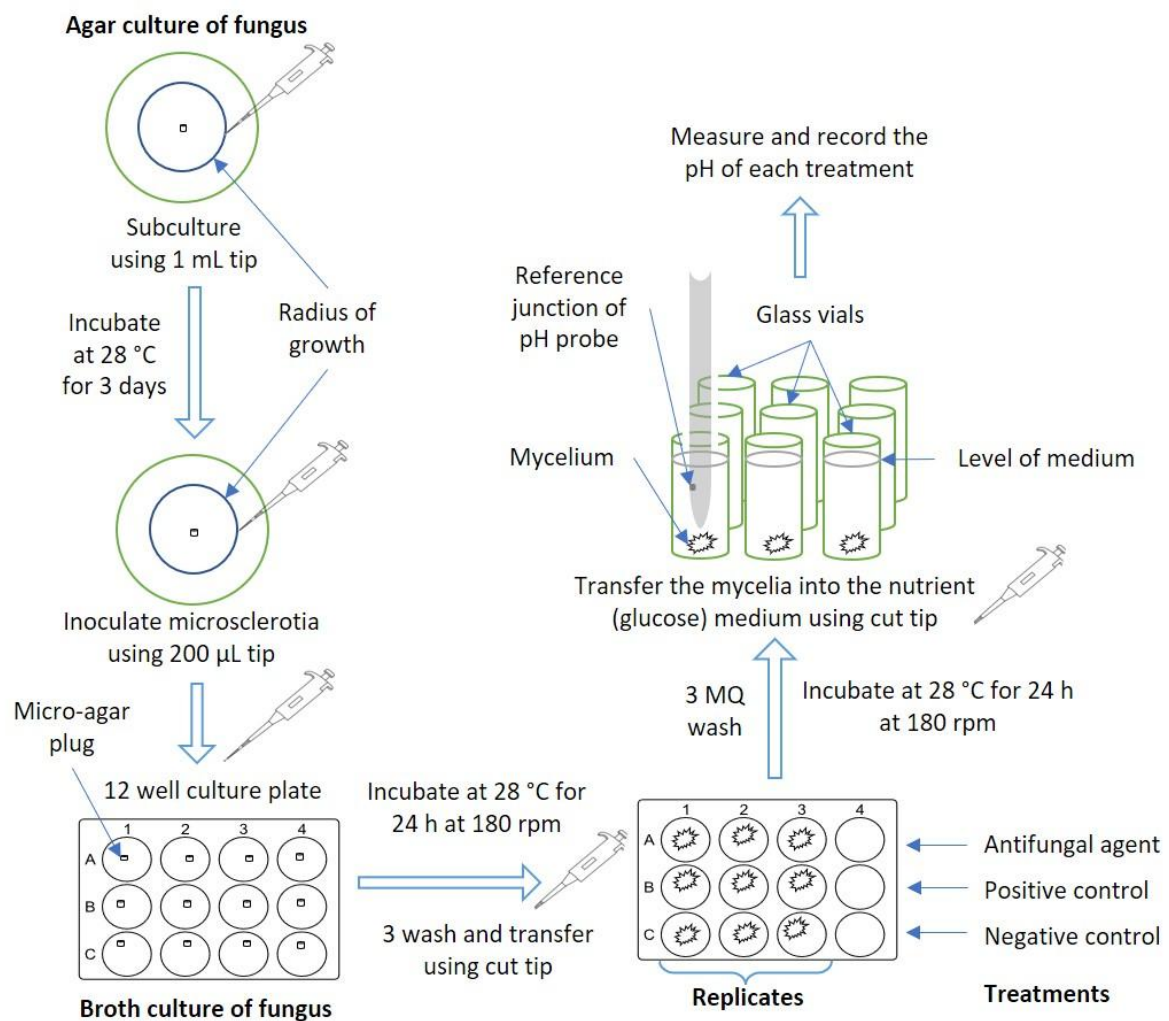
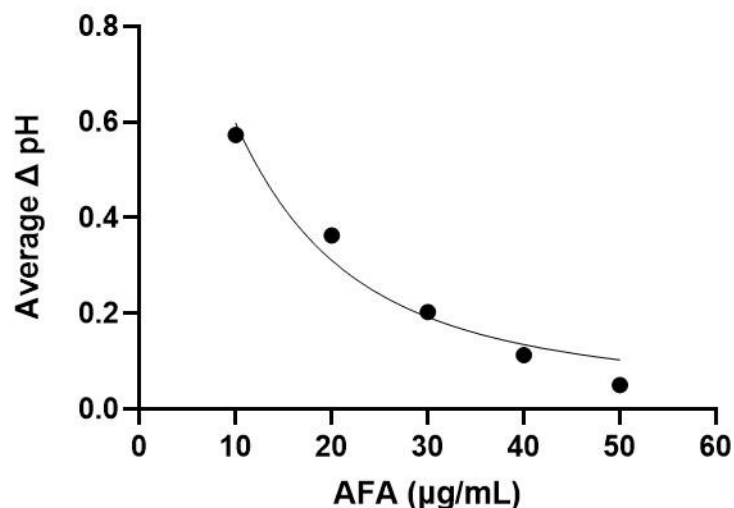


Figure 2. Extracellular acidification assay protocol. The uniformity of agar culture ensures an equal size of micro-agar plug inoculum during subcultures. The microsclerotia collected from a fixed radius of the agar culture are allowed to germinate in the subsequent broth culture. The washing step before and after the treatments removes undesired residual matter, and the cut tips ensure the mycelial integrity



between transfers. The mycelium is suspended in an appropriate medium for measuring the extent of acidification using a pH meter.

MQ: Milli Q water, rpm: Revolutions per minute

Figure 3. Dose-response curve. The curve represents the average of the $\Delta \text{pH}_{\text{initial-final}}$ values for each treatment i.e., the extent of acidification as the response to the individual concentrations of AFA used in the assay. The variable slope model used here fits the Hill slope from the data provided, rather than assuming a standard slope. The software output determines the IC_{50} of the AFA, which is the concentration that elicits a response halfway between the maximum response (Top) of the negative control and the maximum inhibition response (Bottom) of the positive control.

AFA: Antifungal agent, IC_{50} : Inhibitory concentration showing 50% of response